Fiscal Year:	FY 2007	**Task Last Updated:**	FY 08/07/2007
PI Name:	Mansur, Louis K. Ph.D.		
Project Title:	Materials Science and Particle Transport Assessment of Novel and Multifunctional Shielding Materials		
Division Name:	Physical Sciences		
Program/Discipline:	RADIATION SHIELDING		
Program/Discipline--Element/Subdiscipline:	RADIATION SHIELDING--Radiation shielding		
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Space Biology Element:	None		
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Zip Code:	37831-6138	**Congressional District:**	3
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Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
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No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
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Task Description:	The identification, development and qualification of new and multifunctional shielding materials are of the high importance to future manned NASA space missions. Knowledgeable researchers understand that new shielding materials must be developed. Without better shielding for spacecraft crew, long term deep space missions would not be possible. Based on new understanding it is now clear that only those shielding materials that contain high proportions of hydrogen and other light elements, and with an absolute minimum of heavy elements, can be effective against galactic cosmic rays. It is a disturbing realization that recent transport calculations for galactic cosmic rays (GCR), coupled with currently proposed calculational models of biological effects have led to an understanding that conventional spacecraft materials like aluminum alloys may result in radiation effects that are more detrimental than would result with no shielding at all. This effect occurs through fragmentation of the incoming cosmic rays and the heavy nuclei of the spacecraft material such that the sum of the biological damage from the resulting new charged particles, neutrons and energetic photons exceeds that of the original cosmic ray particle. Conventional models of biological effect are based on risk assessment dictated by the magnitude of linear energy transfer (LET) of a particular particle at a given energy, coupled with accepted quality factors that calibrate LET in terms of relative biological effectiveness (RBE) in relation to X-rays (ICRP, 1991). It can be characterized as a dose equivalent method. While this approach may often be useful as a first approximation, it cannot recognize particular mechanisms of action of radiation on organisms. Heavy charged particles may interact with cellular molecules and subassemblies in many different ways, producing detrimental effects that are physically inaccessible to X-rays no matter how high the X-ray dose. An example based on radiation damage physics even without considering a detailed biological pathway is that charged particles with energies as low as a few keV may displace atoms from their molecular sites or cellular subassembly sites to locations far removed, in addition to producing ionization and excitation. X-rays on the other hand cannot generally displace atoms. They produce ionization and excitation. [For a photon to produce displacements to a measurable degree, it generally must be in the energy range of gamma rays above 2 MeV (L. K. Mansur and K. Farrell, J. Nucl. Mater. 212-218 (1997)). While the gamma ray does not produce displacements directly, the resulting Compton electrons and electron/positron pairs can have enough energy to displace atoms.] Thus, there are compelling physical reasons on the most basic level that heavy charged particles may have different and more severe effects than would be obtained by any multiple of X-ray dose. Our approach consists of 1.) Specially preparing advanced shielding materials, 2.) Characterizing their structure and properties, 3.) Carrying out GCR transport and beam transport calculations for both pre-experiment planning and post-measurement analyses, and 4.) Measuring particle types and energy spectra during beam irradiation experiments at the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory, and at the Heavy-ion Medical Accelerator (HIMAC) in Chiba, Japan; analyzing the measured results. In-depth assessments have been carried out with respect to physical, mechanical and shielding properties. Properties such as structural strength, hydrogen storage, and potential for thermal management were assessed on a material specific basis. Calculations of shielding effectiveness will be accompanied by high energy accelerator experiments. A primary aim of these experiments was to determine shielding effectiveness. An equally important goal was to measure the response of the materials to environmental variables. For example, thermal transport and mechanical properties were measured for the carbon materials infiltrated with polyethylene. This research carried out new work on space radiation shielding materials. Both novel shielding materials and multifunctional spacecraft materials that incorporate desirable shielding properties were investigated. The aim was to clearly establish the most promising materials and to assess their key properties with respect to space radiation shielding and suitability for other spacecraft applications. The approach emphasized materials science. The required nuclear properties, in particular the overriding need to incorporate the highest possible proportion of light nuclei, especially hydrogen, were key considerations in selecting and fabricating the experimental materials. The ongoing results of the materials fabrication and testing, particle transport calculations, accelerator based shielding measurements, and associated analyses have been documented in the form of annual progress reports and a final report to NASA. The complete results from this project are available in open literature papers that either have been published or are in press. References to these papers are given in the Bibs, Patents and Awards section of this Task Book Report. (Note that this task was originally scheduled for full funding of $1,180,000 with an ending date of October 7, 2007. However, due to funding cutbacks in the overall shielding budget , the funding and end date of this task were changed by NASA. The revised funding is $909,583 and the revised end date is January 31, 2007)
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Exploration in deep space and missions to Mars are expected to have major benefits on earth. By better understanding the solar system radiation environment and by developing new technologies to protect and support life in hostile conditions, the quality of life on earth with respect to radiation protection can be improved. New materials are necessary to shield spacecraft crew from potentially lethal doses of radiation from galactic cosmic rays. In turn these materials will 1.) bring the benefits to earth that are the outcome of successful missions to space, and 2.) provide an important new suite of materials with novel and multifunctional properties to engineers and designers for earthbound applications.
	Cumulative Final Report for Project, including Progress: September 30, 2006 through January 31, 2007 Mechanical and Physical Properties of New Shielding Materials Work was completed on analyzing the mechanical properties of PE-infiltrated carbon foams and PE-infiltrated carbon fiber monoliths. It was observed that the bending strength of non infiltrated carbon foams increased slightly with foam density. The bending strength of infiltrated foams decreased with increasing foam bulk density, reflecting the effect of a lower percentage of HDPE in the composite as foam density increased. Foam initially of 0.16 g/cm3 density had approximately 88 % by volume occupied by HDPE after infiltration, and this dominated the behavior of the composite. Figures and tables showing these results are available in the papers listed in the Bibs, Patents and Awards section of this report. High Energy Accelerator Based Measurements

Task Progress:

Accelerator irradiations were performed on our materials for examination of their particle shielding properties. At the NSRL all of the 13 (12 candidate materials plus PE control material) materials that we prepared had been irradiated in 1 GeV/u or 0.6 GeV/u 16-O ion beams. The materials were irradiated at nominal thicknesses of ~ 5 g/cm^2. In addition, a thickness study was carried out at additional nominal thicknesses of ~ 1.6, 10 and 15 for one set of materials. Additional runs also were carried out on two materials at nominal thicknesses of ~ 1.6. At the HIMAC six materials, one of which was the PE control material, were irradiated with the 0.65 GeV/u 40Ar beam at nominal thicknesses of ~ 5 g/cm^2. All of the experimental runs were simulated using HETC-HEDS and PHITS high energy particle transport codes. Transmitted primary beam and fragmentation particles were the primary quantities measured as functions of particle beam, energy, specimen thickness and specimen material.

Shielding Calculations with HETC-HEDS

Calculations were performed to simulate our 16-O and 40-Ar shielding experiments at the NSRL and HIMAC. In addition, beams of 16-O at 600 MeV/u and 1 GeV/u and 40-Ar at 650 MeV/u were simulated using the BBFRAG code, which was initially used prior to the experiments to help set up the runs.

Shielding Calculations with PHITS

As in the previous section, computer simulations of the 16-O-ion fragmentation experiments at 0.6 and 1 GeV/u, as well as 40-Ar at 650 MeV/u were performed with the PHITS code. This code offers another method for calculating the shielding fragmentation results using several different physical models and methods of tracking results.

Shielding Calculation Comparisons with Measurements

Studies of results from comparisons of the calculations themselves and for comparisons with experimental data have shown that there is broad agreement between the calculations and the measured experimental data. In finer detail there are differences in the calculated results and in comparison to the measurements. For example, the simulations and experiments showed good agreement for the ions with Z > ~ 8. For lighter ions the differences are considerable. However; it should be noted that the differences in light ion fluxes are not important for the determination of the dose equivalent, because the dose equivalent is dominated by the heavy fragments and, in particular, by the flux of unfragmented (transmitted) initial beam ions. Physical reasons for some of these differences have been identified, and recommendations for future work to pin down the bases for other differences have been given. All of these results are available in the papers listed in the Bibs, Patents and Awards section of this report.

Progress: September 1, 2005 through September 30, 2006

Mechanical and Physical Properties of New Shielding Materials

Work continued on analyzing the mechanical properties of PE-infiltrated carbon foams and PE-infiltrated carbon fiber monoliths. It was observed that the bending strength of non infiltrated carbon foams increased slightly with foam density. The bending strength of infiltrated foams decreased with increasing foam bulk density, reflecting the effect of a lower percentage of HDPE in the composite as foam density increased. Foam initially of 0.16 g/cm3 density had approximately 88 % by volume occupied by HDPE after infiltration, and this dominated the behavior of the composite. A figure showing these results is available in the pdf version of this report.

Work is also ongoing on the preparation of three journal articles reporting our results. The first one will cover the properties of foams and PE-infiltrated foams, the second will cover the corresponding results for the carbon fiber monoliths, while the third one will cover our results on the methane storage capacity of activated carbons as a function of temperature.

Shielding Calculations with HETC-HEDS

Calculations were performed to simulate our 16-O and 40-Ar shielding experiments that were carried out at the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory and at the HIMAC accelerator in Japan, respectively.

As described in the earlier progress report of 2005, and reviewed in this paragraph for context, accelerator irradiations were performed on a variety of high hydrogen containing materials for examination of their particle shielding properties. At the NSRL all of the 13 (12 candidate materials plus PE control material) materials that we prepared had been irradiated in 1 GeV/u or 0.6 GeV/u 16-O ion beams. The materials were irradiated at nominal thicknesses of ~ 5 g/cm^2. In addition, a thickness study was carried out at additional nominal thicknesses of ~ 1.6, 10 and 15 for one set of materials. Additional runs also were carried out on two materials at nominal thicknesses of ~ 1.6. All of the experimental runs were simulated in the present calculations using HETC-HEDS (High Energy Transport Code - Human Exploration and Development of Space).

At the HIMAC six materials, one of which was the PE control material, were irradiated with the 0.65 GeV/u 40Ar beam at nominal thicknesses of ~ 5 g/cm^2. HETC-HEDS simulations were carried out for each of these runs also. The compositions of the materials are contained in a table in the pdf version of this report..

In addition, beams of 16-O at 600 MeV/u and 1 GeV/u and 40-Ar at 650 MeV/u were simulated using the BBFRAG code, which was initially used prior to the experiments to help set up the runs. The purposes of all these computational simulations were to investigate shielding properties for various candidate materials to be used for shielding astronauts on future lunar and interplanetary missions from galactic cosmic rays, as well as to validate the codes against experimental data.

The simulations tracked only the primary particles from the heavy ion beams. In all but one case, we simulated the paths of 10^6 particles through the specimens as well as the particle detectors in the experimental setup. The resulting data files were then processed using locally written FORTRAN programs to find the energy deposited in the target material in each run as well as the particle fluences of all important ions of interest entering and leaving the target material. The programs were compiled and run on three systems at the University of Tennessee: a pair of dedicated computational machines and a desktop computer. A figure in the pdf version of this report shows, for example, calculated simulations of all of the 40-Ar runs. All of the calculated results are available in a full set of Microsoft Excel spreadsheets.

Shielding Calculations with PHITS and Comparison with Measurements

In support of the development and characterization of new shielding materials as in the previous section, computer

simulations of the 16-O-ion fragmentation experiments at 0.6 and 1 GeV/u, as well as 40-Ar at 650 MeV/u were performed with the PHITS code. This code offers another method for calculating the shielding fragmentation results. Studies of results from comparisons of the calculations themselves and for comparisons with experimental data are underway.

PHITS [Iwase, et al., 2002; Iwase, 2003] is a three dimensional radiation transport code system with the capabilities of transporting heavy ions as well as protons, neutrons, charged pions, electrons, positrons, gamma rays, etc. The main transport module and the proton, neutron, and charged pion collision modules of PHITS were obtained originally from the ORNL code NMTC [Bertini, 1963] and have been modified with various upgrades and improvements. Jet AA Microscopic Transport Model (JAM) [Nara, et al., 1999] and JAERI Quantum Molecular Dynamics (JQMD) [Niita, et al., 2001] are used to simulate particle-induced reactions up to about 200 GeV, and nucleus–nucleus collisions up to about 100 GeV, respectively. Fission and evaporation processes are treated by the Generalized Evaporation Model (GEM) [Furihata, 2000]. The GEM code calculates the final yields of light nuclei produced from the de-excitation of residual nuclei. Neutrons that emerge from the nucleus below selected threshold (e.g, 20 MeV) can then be transported using the MCNP4C radiation transport code [Briemeister, 1997], which is incorporated in PHITS. For the below-threshold neutron transport PHITS employs the cross sections from evaluated nuclear data libraries such as ENDF-B/VI and JENDL-3.3 up to 20 MeV and LA150 up to 150 MeV. The geometry input to PHITS may be either Combinatorial Geometry (CG) or Generalized Geometry (GG).

While PHITS offers state-of-the-art methods for ion transport calculations, the JQMD module requires substantial computational capabilities and the simulations were performed on the Beowulf cluster at Oak Ridge National Laboratory, with the MPI version of PHITS running in parallel. A typical run simulated about 106 source particles and required from one to several days of CPU time.

Preliminary results from the present work were reported at the ANS Radiation Protection and Shielding topical Meeting in Carlsbad, NM in April 2006 [Remec, et al., 2006]. An additional example of simulation results to those contained in that presentation is given in a figure available in the pdf version of this report, which shows fragmentation of 40Ar ions at 650MeV/u on a low density polyethylene target (LDPE) at ~ 5.4 g/cm^2 thickness. The experimental results had been corrected for the effects of beam fragmentation in the detectors and in the air, so these effects were not considered in the calculations. The simulations modeled the beam-target-detector geometry experimental setup. The flux of fragments was tallied directly using volume averaged and surface crossing flux tallies. The results of the simulation performed in this way is labeled as “Calc. F-Current” in the figure. This simulation tallies ions passing through the detector volume and/or surface individually. If several fragments pass through the detector simultaneously, they contribute to all appropriate fragment tallies. However, the experiment was not able to distinguish between near-simultaneous crossings of several particles, such as occurs if several fragments from the same ion-projectile hit the detector. The experiment registered such an event as a one-particle crossing with the charge number deduced from the combined energy deposition of all the fragments. To simulate this effect an improved simulation was performed in which all the fragments from the same projectile that hit the detector were counted as one particle only, and the particle’s effective charge was determined as the square root of the sum of squares of the charges of all the fragments. The result of this simulation is labeled as “Modified Calc.” in the figure. As expected the two methods of simulation produced practically identical results for higher Z ions, however the improved simulation method agreed much better with the experimental results for the proton fluxes (Z=1).

The simulations and experiment show good agreement for the ions with Z > ~ 8. For lighter ions the differences are considerable. However; it should be noted that the differences in light ion fluxes are not important for the determination of the dose equivalent, because the dose equivalent is dominated by the heavy fragments and, in particular, by the flux of unfragmented (transmitted) initial beam ions.

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